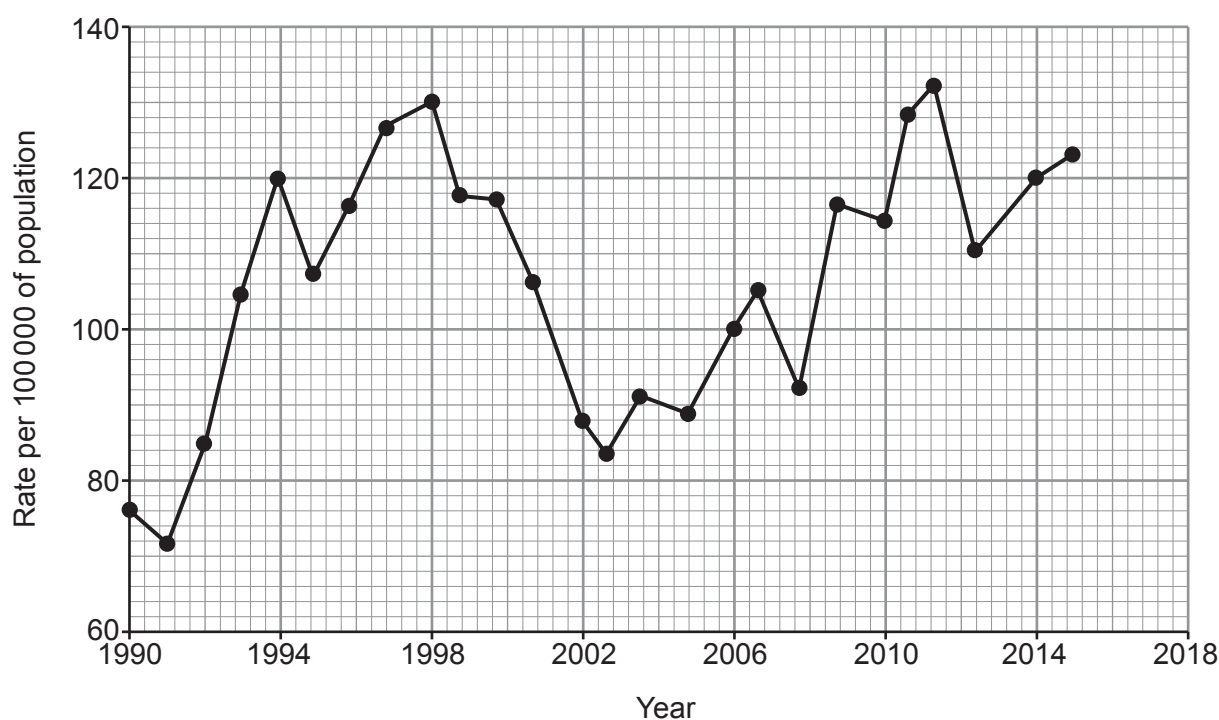


8. The graph below shows the confirmed cases of *Campylobacter* food poisoning per 100 000 of the population in Wales from 1990–2015.



In Wales, for every confirmed case of *Campylobacter* poisoning there are another two unconfirmed cases. The population of Wales is 3 million.

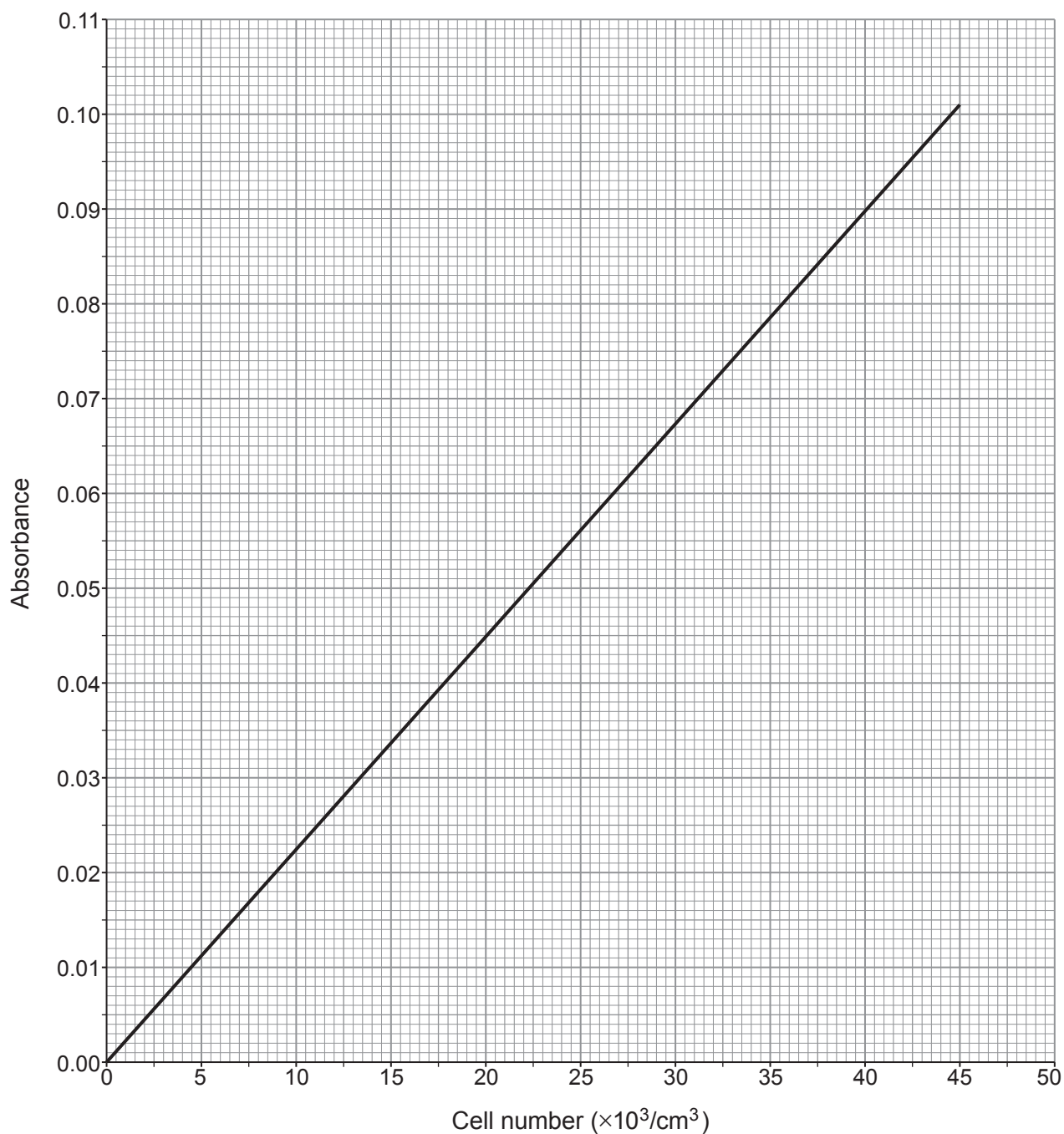
- (a) Calculate the total number of *Campylobacter* cases in Wales in 1998.

[3]

total number of cases =



- (b) Turbidity can be used to calculate the number of bacteria present in a sample. Alun used turbidity to study the growth rate of *Campylobacter* in water. He used the following standard graph.



(i) Alun analysed a water sample. The sample gave an absorbance of 0.01.

I. Use the graph to find the cell number per cm^3 at an absorbance of 0.01. [1]

cell number = per cm^3

II. *Campylobacter* reproduce every 20 mins The sample was left to stand for 1 hour. Alun thought that the absorbance would be 0.03 after 1 hour. Use your knowledge and data in the graph to explain whether he is correct. [3]

Space for calculations.

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(ii) The threshold value for making a person sick is 120 000 bacteria. In (b)(i)I. Alun's sample was 7 cm^3 . Use your answer above to determine whether this sample of water would make a person sick after standing for the hour. [3]

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END OF PAPER





THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																	
7 Li Lithium 3	9 Be Beryllium 4																
23 Na Sodium 11	24 Mg Magnesium 12																
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number